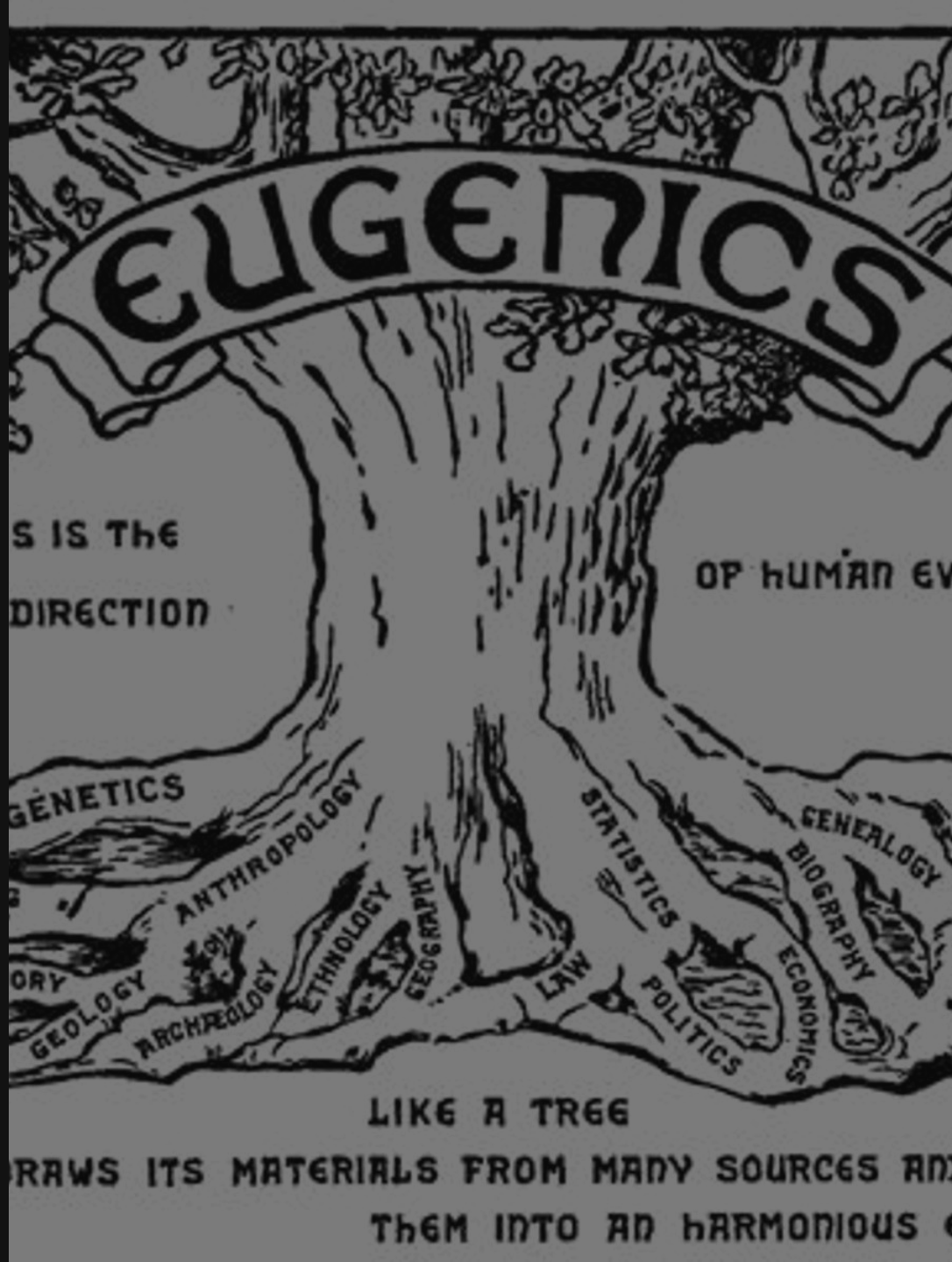
What's Wrong With Science?

The Dark Side of Science

Harm · Doubt · Ethics · Safeguards

Session 4 · Professor David Gresham · Fall 2026

Guest: Nina Gray, NYU Interim Vice Provost for Research





Dr Nina Gray, NYU Interim Vice Provost for Research

Before we start

Tonight includes non-consensual research, racism, and forced sterilization.

It's okay to step out. Wellness Exchange: 212-443-9999

How do we decide what science gets done and how do we ensure that it is done ethically?

Organization

Institutional Review Board

Institutional Review Board (IRB)

The Institutional Review Board (IRB) reviews human subjects research studies conducted under the auspices of NYU Washington Square. It is comprised of faculty and staff from throughout the University and is supported by the IRB Office. The IRB Office assists researchers to obtain IRB approval and to ensure that their research is compliant with applicable regulations/laws and with University policies/guidance.



You Are the Ethics Board

Groups: combine odd and even row starting at row 1· read research proposals · 20 minutes

1

Read — 3 min

Six short proposals, A–H

2

Discuss — 9 min

Approve · Approve with changes · Reject

3

Vote — 3 min

Vote on Poll Everywhere

Judge each proposal as if it landed on your desk today.

You Are the Ethics Board: Proposal A-H



- Approve
- Approve with changes
- Reject

PROPOSAL



To study social media, university researchers and a social media company will change the news feeds of about 690,000 users for one week, showing some fewer positive posts and others fewer negative posts. They will then measure the emotional tone of what those users post. Users will not be told about the study as all users agreed to the company's data-use policy when they signed up.

Approve

Approve with changes

Reject

Proposal Status

Approve

0%

Approve with changes

0%

Reject

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

PROPOSAL

B

A research institute will train field workers to visit prisons, hospitals, and economically disadvantaged families and record family histories of important social traits such as intelligence, work ethic, and criminal records. Pedigree analysis will be performed to determine which traits are inherited. Findings will inform lawmakers on marriage, immigration, and fertility policies.

Approve

Approve with changes

Reject

Proposal Status

Approve

0%

Approve with changes

0%

Reject

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

PROPOSAL

C

Researchers will deliberately infect 36 healthy volunteers, aged 18–30, with a pandemic virus to find the smallest dose that causes infection. This will speed up the testing of vaccines and treatments to address a global crisis. Volunteers are fully informed, paid £4,500, and quarantined in hospital for two weeks, with treatment on hand if they become ill.

Approve

Approve with changes

Reject

Proposal Status

Approve

0%

Approve with changes

0%

Reject

0%

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PROPOSAL

D

To develop a new weapon workers need to handle a newly isolated radioactive element. To set safe exposure limits, scientists need to understand how the body absorbs and excretes it. Researchers will inject small doses into hospital patients who are terminally ill and monitor lifetime of the isotope in the human body by collecting their urine. To ensure rigor and minimize the placebo effect patients will not be informed.

Approve

Approve with changes

Reject

Proposal Status

Approve

0%

Approve with changes

0%

Reject

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

PROPOSAL

E

CRISPR genome-editing will be used to edit embryos from couples in which the father has HIV, disabling a gene encoding a protein that the virus uses to enter cells. The edited embryos will be implanted and the children followed for 18 years. Detailed consent forms are provided for this AIDS vaccine development project.

Approve

Approve with changes

Reject

Proposal Status

Approve

0%

Approve with changes

0%

Reject

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

PROPOSAL

F

A government health service will study about 600 African American men, 400 with syphilis and 200 without, to determine how untreated syphilis progresses over a lifetime. The participants will receive free medical exams and hot meals.

Approve

Approve with changes

Reject

Proposal Status

(A) Approve

0%

(B) Approve with changes

0%

(C) Reject

0%

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PROPOSAL

G

Researchers will randomly assign 1,140 pregnant women who smoke to receive either nicotine patches or e-cigarettes, to see which better helps them quit. Long-term effects of vaping on a developing fetus are unknown, but continued smoking is known to harm it. Women are fully informed and can stop at any time.

Approve

Approve with changes

Reject

Proposal Status

Approve

0%

Approve with changes

0%

Reject

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

PROPOSAL

H

Leftover tissue from a patient’s tumor biopsy, taken during routine cancer treatment, will be used to grow human cells that survive indefinitely in the lab. The cells will be shared with laboratories around the world to enable biomedical research. The patient is now deceased. Using leftover tissue without consent is standard practice at hospitals.

Approve

Approve with changes

Reject

Proposal Status

Approve

0%

Approve with changes

0%

Reject

0%

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Each of these events and/or research projects happened IRL.



RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES |

Experimental evidence of massive-scale emotional contagion through social networks

Adam D. I. Kramer, Jamie E. Guillory, and Jeffrey T. Hancock Authors Info & Affiliations

Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved March 25, 2014 (received for review October 23, 2013)

June 2, 2014 | 111 (24) 8788-8790 | https://doi.org/10.1073/pnas.1320040111

THIS ARTICLE HAS BEEN CORRECTED + THIS ARTICLE HAS AN EXPRESSION OF CONCERN +

2,051,111 | 2,301 PDF

Significance

We show, via a massive (*N* = 689,003) experiment on Facebook, that emotional states can be transferred to others via emotional contagion, leading people to experience the same emotions without their awareness. We provide experimental evidence that emotional contagion occurs without direct interaction between people (exposure to a friend expressing an emotion is sufficient), and in the complete absence of nonverbal cues.

PROPOSAL A

The Facebook “emotional contagion” study

2012 · Facebook & Cornell

- Ran without users’ knowledge; published in PNAS, 2014
- Cornell’s ethics board said no review was needed, because Facebook ran the experiment
- Public outcry; the journal issued an “Expression of Concern”



PROPOSAL B

The Eugenics Record Office

1910–1939 · Cold Spring Harbor, NY

- Funded by the Carnegie Institution
- Its family studies were built on hearsay and social prejudice
- Its data supported sterilization laws in 30+ states and the 1924 Immigration Act

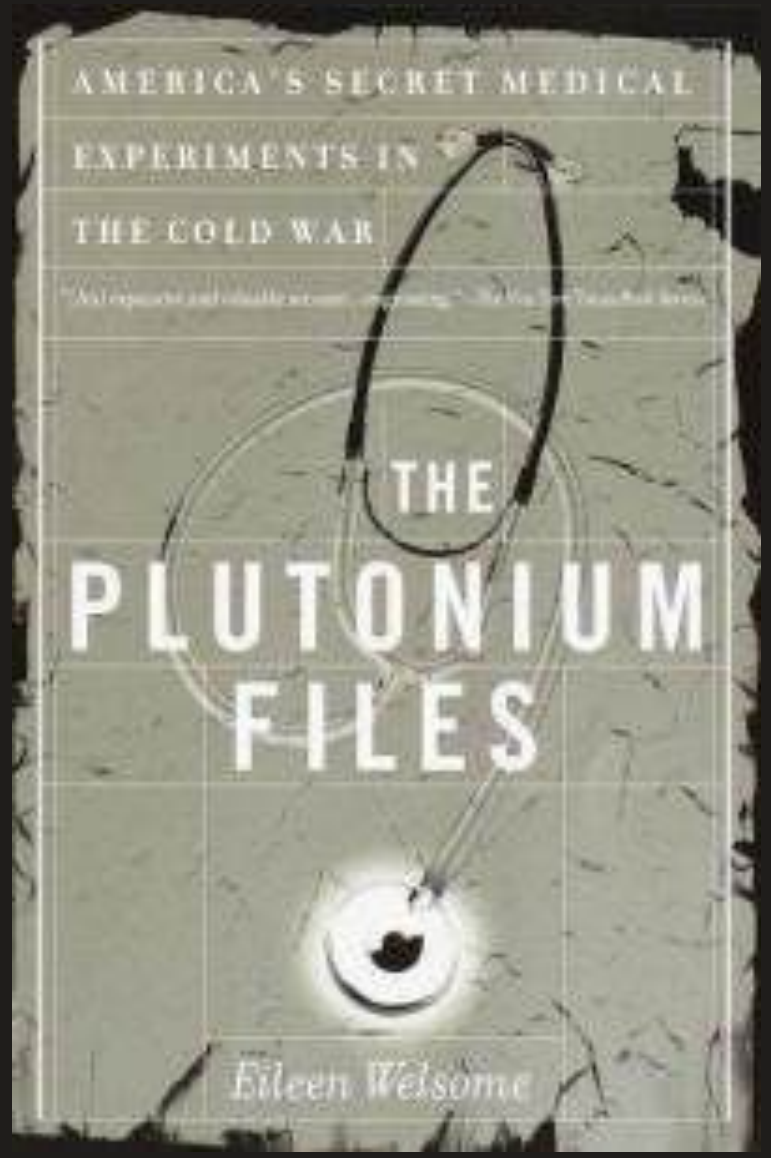


PROPOSAL C

UK COVID-19 human challenge trial

2021 · London

- Approved by a UK ethics committee
- About half the volunteers became infected
- No one became seriously ill



PROPOSAL D

The Manhattan Project plutonium injections

1945–1947 · U.S. hospitals

- 18 patients injected with plutonium without their knowledge
- Several were not terminally ill; one man lived 20 more years
- Exposed by journalist Eileen Welsome in 1993; U.S. government apology, 1995



PROPOSAL E

He Jiankui and the CRISPR babies

2017–2018 · Shenzhen, China

- Three children born with edited genomes
- Forged ethics approval
- Safe ways to prevent HIV transmission already existed
- Sentenced to 3 years in prison



PROPOSAL F

The Tuskegee syphilis study

1932–1972 · U.S. Public Health Service

- Ran for 40 years; penicillin withheld after it became the cure
- Published in medical journals; ended only by a whistleblower
- Led directly to the Belmont Report and modern ethics boards



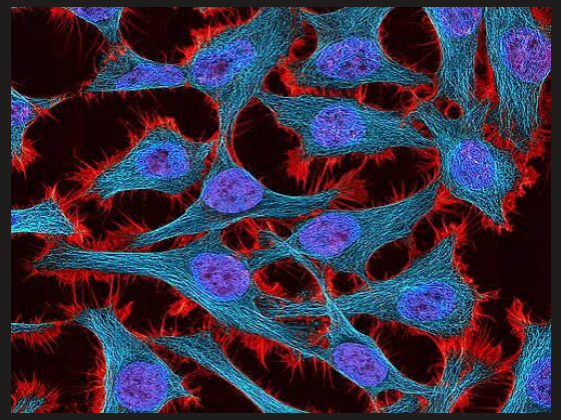
PROPOSAL G

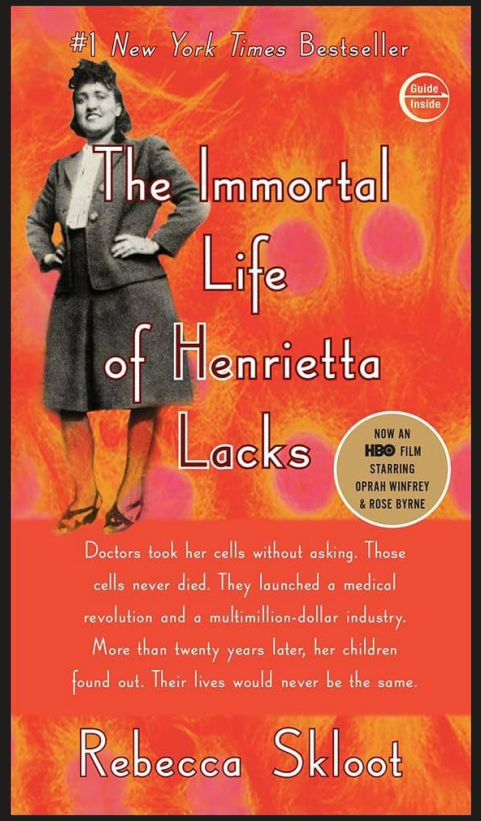
E-cigarettes vs. nicotine patches in pregnancy

Published 2022 · 23 hospitals in England, plus Scotland

- 1,140 pregnant smokers; funded by NIHR and Public Health England
- Safety similar in both groups; low birthweight less common with e-cigarettes







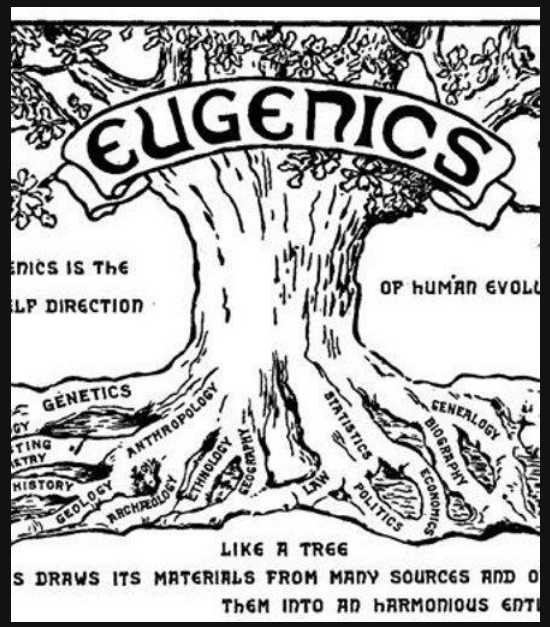
PROPOSAL H

HeLa cells: Henrietta Lacks

1951 · Johns Hopkins Hospital, Baltimore

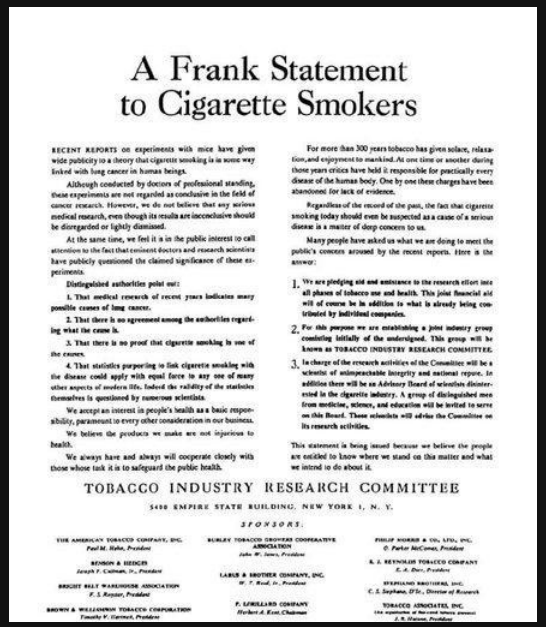
- Cells taken from a 31-year-old woman’s tumor without her knowledge
- Basis of the polio vaccine, cancer research, and IVF;
- Sold commercially for decades
- Her family learned 20+ years later

Four questions



1 Harm

When and how has science itself done harm?



A Frank Statement to Cigarette Smokers

RECENT REPORTS on experiments with mice have given wide publicity to a theory that cigarette smoking is in some way linked with lung cancer in human beings.

Although conducted by doctors of professional standing, these experiments are not regarded as conclusive in the field of cancer research. However, we do not believe that any serious medical research, even though its results are inconclusive should be disregarded or lightly dismissed.

At the same time, we feel it is in the public interest to call attention to the fact that eminent doctors and research scientists have publicly questioned the claimed significance of these experiments.

Distinguished authorities point out:

1. That medical research of recent years indicates many possible causes of lung cancer.

2. That there is no agreement among the authorities regarding what the cause is.

3. That there is no proof that cigarette smoking is one of the causes.

4. That statistics purporting to link cigarette smoking with the disease could apply with equal force to any one of many other aspects of modern life. Indeed the validity of the statistics themselves is questioned by numerous scientists.

We accept an interest in people's health as a basic responsibility, paramount to every other consideration in our business.

We believe the products we make are not injurious to health.

We always have and always will cooperate closely with those whose task it is to safeguard the public health.

For more than 300 years tobacco has given solace, relaxation, and enjoyment to mankind. At one time or another during those years critics have held it responsible for practically every disease of the human body. One by one these charges have been abandoned for lack of evidence.

Regardless of the record of the past, the fact that cigarette smoking today should even be suspected as a cause of a serious disease is a matter of deep concern to us.

Many people have asked us what we are doing to meet the public's concern aroused by the recent reports. Here is the answer:

1. We are pledging aid and assistance to the research effort into all phases of tobacco use and health. This joint financial aid will of course be in addition to what is already being contributed by individual companies.

2. For this purpose we are establishing a joint industry group consisting initially of the undersigned. This group will be known as TOBACCO INDUSTRY RESEARCH COMMITTEE.

3. In charge of the research activities of the Committee will be a scientific of unimpeachable integrity and national repute. In addition there will be an Advisory Board of scientists disinterested in the cigarette industry. A group of distinguished men from medicine, science, and education will be invited to serve on this Board. These scientists will advise the Committee on its research activities.

This statement is being issued because we believe the people are entitled to know where we stand on this matter and what we intend to do about it.

TOBACCO INDUSTRY RESEARCH COMMITTEE

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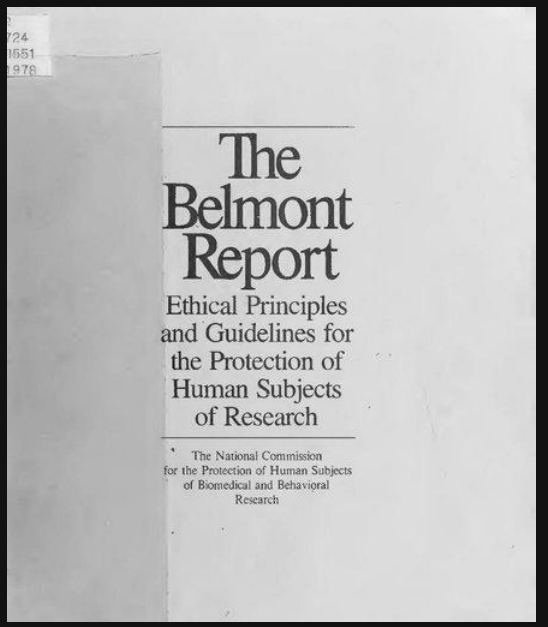
2 Doubt

What is the role of scientists in manufacturing doubt?



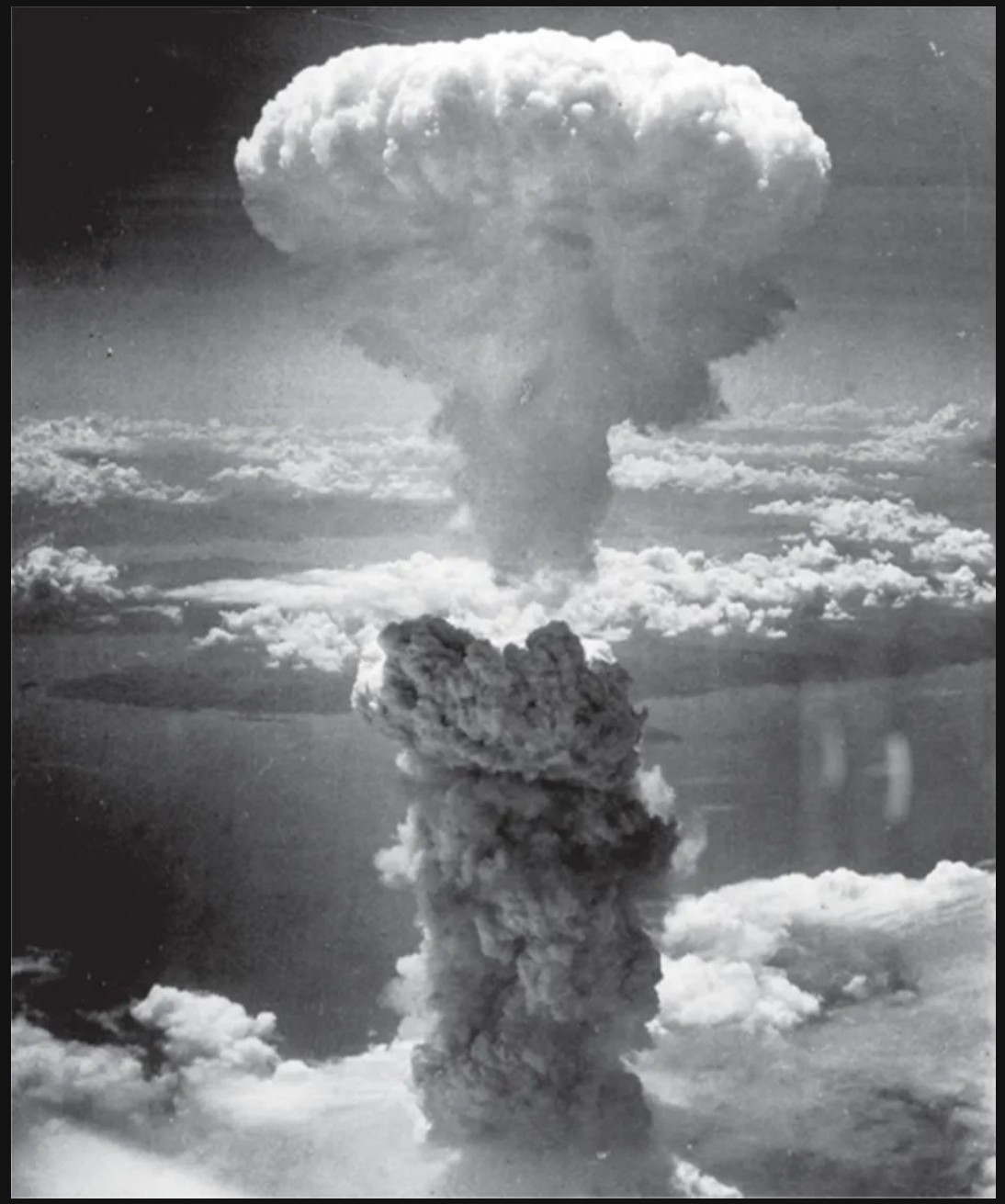
3 Ethics

How is ethical research regulated?



4 Safeguards

What is done in response to scandal?







Now I am become Death, the destroyer of worlds.

-Bhagavad Gita (chapter 11, verse 32)

Top A.I. Leaders Call for Slowing Down A.I. Development

In an essay, the head of Anthropic laid out the rapidly advancing capabilities of artificial intelligence and said there needed to be greater safety controls across the industry.

Listen · 7:38 min

Share full article

1.4K



Dario Amodei, Anthropic's chief executive, in February. "Over the last few months, I have become convinced that fully addressing the risks requires even more prudence," he wrote in an essay. Ludovic Marin/Agence France-Presse — Getty Images

BBC

Stop guessing, get your LLC right the *first time*
Start today

AI staff 'genuinely frightened' for humanity's future, ex-Anthropic researcher tells BBC

13 September 2026

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Brandon Livesay and **Mitchell Labiak,** business reporter



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TheUpshot

A.I. or Nuclear Weapons: Can You Tell These Quotes Apart?

Many experts on artificial intelligence are warning of its potential dangers and calling for regulation, just as others once did with the atomic bomb.

Share full article 225

Can you tell which subject these quotations are about?

1 of 12 "_____ is the future not only of Russia but of all of mankind. There are huge opportunities, but also threats that are difficult to foresee today. Whoever becomes the leader in this sphere will become the ruler of the world."

- Artificial intelligence
- Nuclear technology

2 of 12 "Certainly, measures of control can be devised which will eliminate the fear of sudden, total devastation which is what we're headed for now."

- Artificial intelligence
- Nuclear technology

https://www.nytimes.com/2023/06/10/upshot/artificial-intelligence-nuclear-weapons-quiz.html?smid=url-share

TheUpshot

A.I. or Nuclear Weapons: Can You Tell These Quotes Apart?

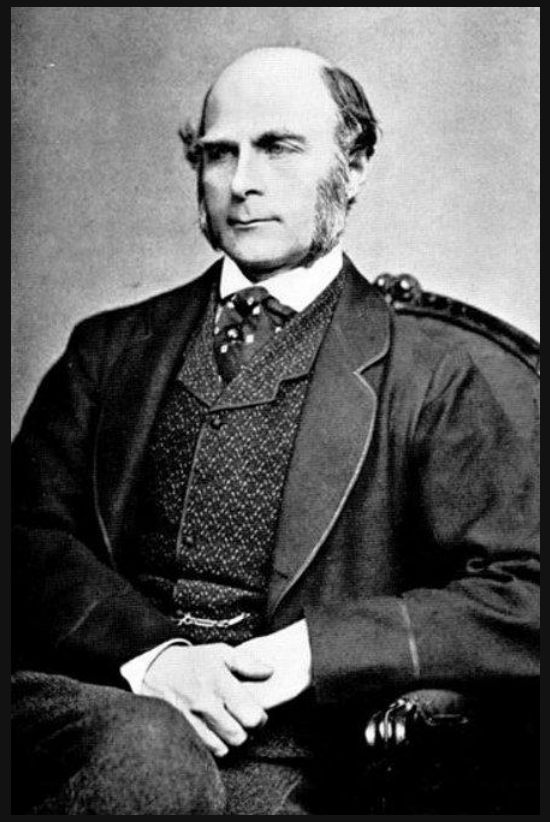
Many experts on artificial intelligence are warning of its potential dangers and calling for regulation, just as others once did with the atomic bomb.

Share full article 225

Score **8 out of 12**

The above quotes are only a slice of the responses to — and the debate over — A.I. and nuclear technology. They capture parallels, but also some notable differences: the fears of imminent, fiery destruction from atomic weapons; or how advancements in A.I. right now are mostly the work of private companies rather than governments.

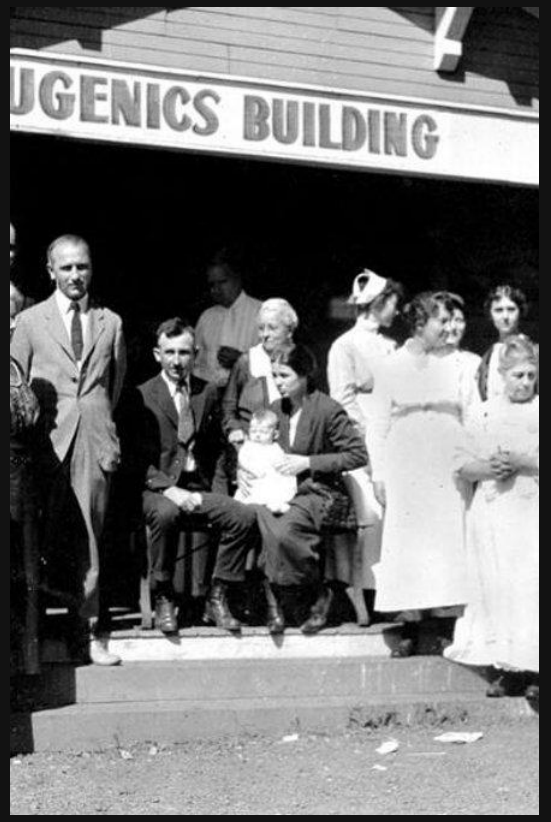
In the early 20th century Eugenics was the scientific consensus



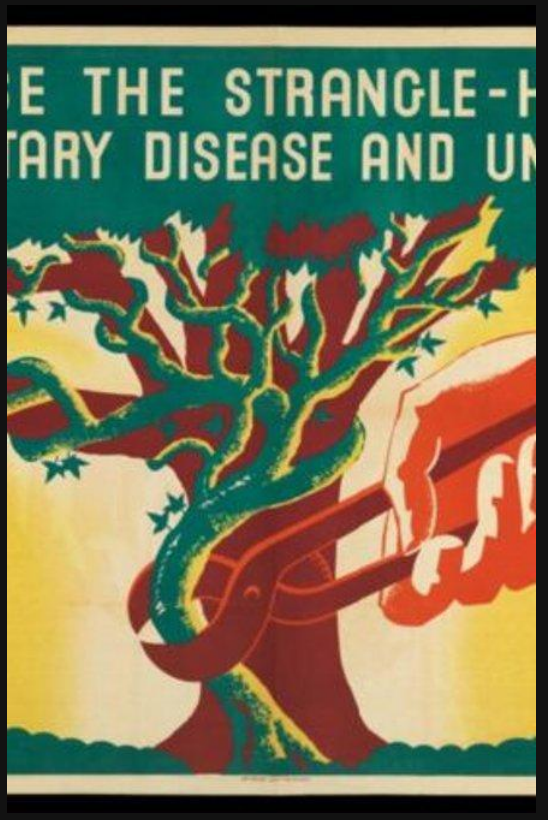
Francis Galton coins “eugenics” (1883)



Eugenics Record Office, Cold Spring Harbor (1912)



“Fitter Families” contest, Kansas State Fair



Eugenics Society poster, 1930s

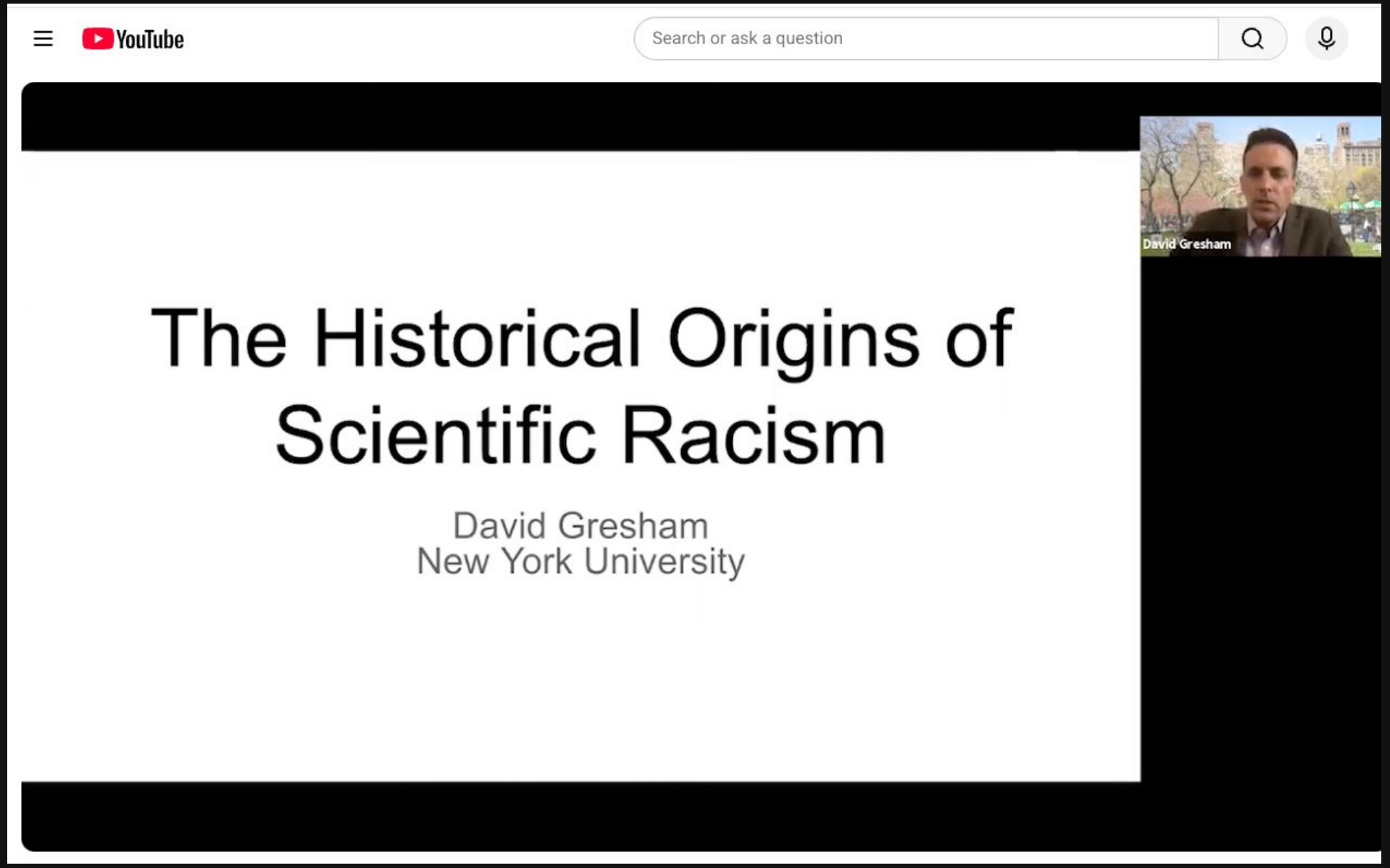
60,000+

Americans forcibly sterilized

32

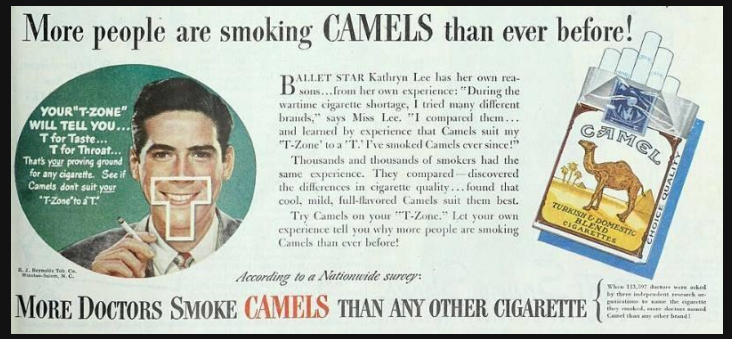
states with sterilization laws



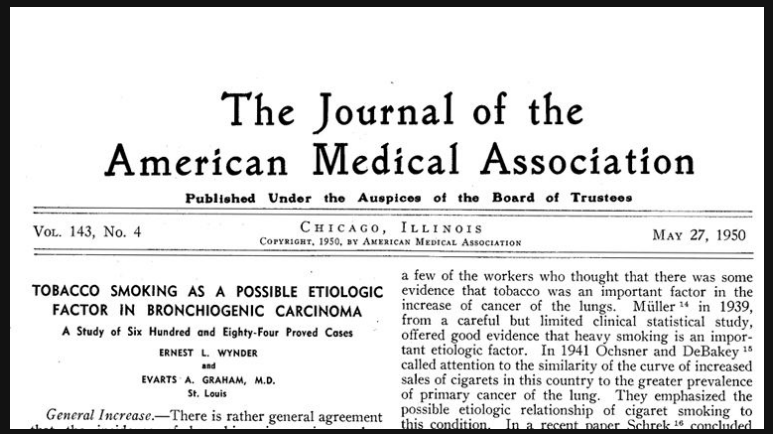


https://www.youtube.com/watch?v=zuNP8ZglS8g

Tobacco: fomenting doubt



1940s



The Journal of the American Medical Association

Published Under the Auspices of the Board of Trustees

VOL. 143, No. 4 — CHICAGO, ILLINOIS — MAY 27, 1950

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TOBACCO SMOKING AS A POSSIBLE ETIOLOGIC FACTOR IN BRONCHIOGENIC CARCINOMA

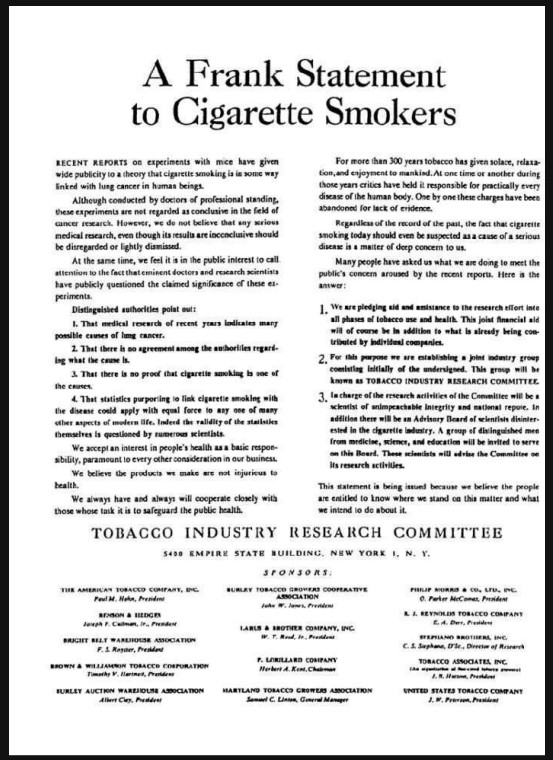
A Study of Six Hundred and Eighty-Four Proved Cases

ERNEST L. WYNDER
and
EVARTS A. GRAHAM, M.D.
St. Louis

General Increase.—There is rather general agreement

a few of the workers who thought that there was some evidence that tobacco was an important factor in the increase of cancer of the lungs. Müller[14] in 1939, from a careful but limited clinical statistical study, offered good evidence that heavy smoking is an important etiologic factor. In 1941 Ochsner and DeBakey[15] called attention to the similarity of the curve of increased sales of cigarets in this country to the greater prevalence of primary cancer of the lung. They emphasized the possible etiologic relationship of cigaret smoking to this condition. In a recent paper Schrek[16] concluded

1950



A Frank Statement to Cigarette Smokers

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3. In charge of the research activities of the Committee will be a scientist of unimpeachable integrity and national repute. In addition there will be an Advisory Board of scientists disinterested in the cigarette industry. A group of distinguished men from medicine, science, and education will be invited to serve on this Board. These scientists will advise the Committee on its research activities.

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1954

“Doubt is our product.”

Brown & Williamson tobacco company, internal memo, 1969



Scientists contributed to creating a culture of doubt about the connection between lung cancer and smoking

Frederick Seitz
Former president, National Academy of Sciences

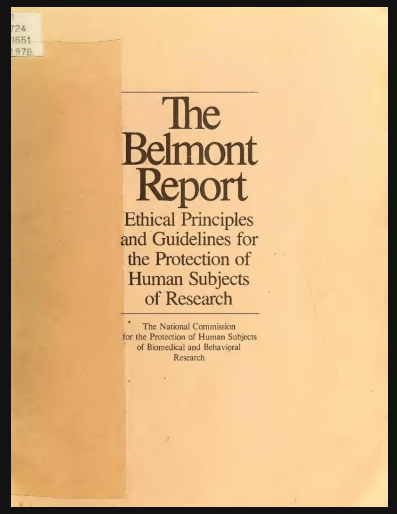
Directed R.J. Reynolds' research funding

Oreskes & Conway, Merchants of Doubt (2010)

Safeguards are established in response to crises and/or whistleblowers



Nuremberg Doctors' Trial, 1946–47



1947

Nuremberg Code

1974

National Research Act: IRBs

1979

Belmont Report

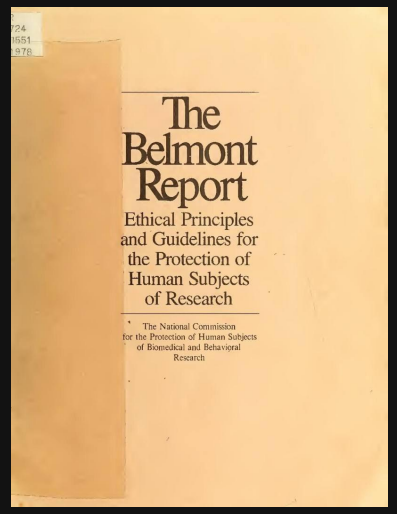
2019

WHO genome-editing registry

Safeguards are established in response to crises



Nuremberg Doctors' Trial, 1946–47



Three principles:

1. **Respect for persons.** People must be treated as able to make their own decisions, which is the basis for (informed consent).
2. **Beneficence.** Do no harm, and make sure the benefits of research outweigh the risks to participants.
3. **Justice.** The burdens and benefits of research must be shared fairly, so vulnerable groups are not used as subjects for research that mainly benefits others. The report cites Tuskegee directly as a violation.